# SPECTRAL SYSTEMS LLC CAGE: 268E1 • SAM.gov UEI: VTP2JPCJU3D9 • VA eVA: VA10140357

Hardware-Enforced CXL 3.0--5.0 / UCIe Memory Isolation & Formal Silicon Verification Web: https://spectralsystems.org • POC: eric@spectralsystems.org

## 1. EXECUTIVE OVERVIEW & CORE CAPABILITY (TRL-6/7 SILICON & TRL-8 CLOUD CONTROL PLANE)

Spectral Systems LLC designs and licenses the **Spectral Flit-Filter (v1.0.0)**, a stateless, cut-through hardware security interposer for disaggregated Compute Express Link (CXL 3.0–5.0), PCIe Gen5/Gen6, and UCIe High-Bandwidth Memory (HBM2e/HBM3/HBM4) fabrics. Developed 100% at private expense (DFARS 252.227-7018 / DEAR 952.227-14; 19 U.S. Provisional Patent Applications, *All Patents Pending*), the core evaluates spatial, temporal, and tenant capability masks across all four 128-bit sub-flit slots of a 512-bit flit in a single 0.124 ns logic level using **zero Block RAM (0 BRAM / 0 URAM / 0 DSP)** while sustaining **99.20% link goodput** (vs. 48.90% under whole-flit CRC poisoning) and reclaiming **2,500 TB of stranded DRAM** ($37.50M–$38.40M CapEx) per 10,000-rack cluster. **2. VERIFIED EMPIRICAL SILICON, ASIC & FORMAL PROOF BASELINE (AUDITABLE IN 60 SECONDS)**

| Verification Track | Platform / Toolchain | Measured Post-Route / Live Hardware Telemetry |
|---|---|---|
| **1. Formal Proof Kernel** | Microsoft Lean 4 (Init.Prelude) | **30 Constructive Theorems** (thm00–thm29, 0 sorry, 0 axioms) down to placed 16nm LUT6_2 truth tables. |
| **2. 16nm FinFET FPGA** | AMD UltraScale+ xcvu47p (Vivado) | **1,585 Core CLB LUTs** (0.12%), **0 BRAM/URAM/DSP**, 0.124 ns Stage-1 sieve, +1.048 ns WNS (338.8 MHz), 35.0 mW. |
| **3. 7nm FinFET CMOS** | ASAP7 7nm PDK (OpenROAD/STA) | **9,678 Core / 27,877 Enclave Cells** (0.003335 $mm^2$ full enclave), **693.19 MHz** Fmax, **3.728 mW**, 0 SRAM. |
| **4. AWS F2 BIST & TVLA** | agfi-03b742e2a9cd2c0f2 (f2.6xlarge) | **800,000 BIST sub-flit slots** @ 250 MHz (128 Gbps), warm PCIe FLR survival (CSR[0x70]=0xFFFFFFFFFFFFFFEB), TVLA max $\|t\|$ = 0.0. |
| **5. AWS F2 16 GB HBM2** | agfi-08becf3b40b169674 (f2.6xlarge) | **100,000 inline DMA slots** across physical 16 GB HBM2 (PF0-BAR4), 100% sentinel survival (0xAAAAAAAAAAAAAAAA), 0 CTOs. |
| **6. Kubernetes Operator** | spectralc + CNCF k3s Webhook | FIPS-204 ML-DSA-87 signed policy bundles; blocks non-monotonic privilege widening (HTTP 403) in **1.36 ms**. |

## 3. KEY DIFFERENTIATORS FOR DOD, NASA, DOE, IC & HYPERSCALE PARTNERS

- **Cold-Power-Isolated Monotonic Ratchet (thm06):** Capability register CSR[0x70] enforces a physical AND2 latch (q <= q & wdata) reset exclusively by cold power-on reset (por_cold_rst_n). Even a Ring-0 host root compromise + warm PCIe reset cannot widen memory permissions.
- **Radiation-Resilient 0-BRAM Footprint (DO-254 DAL-A & NSA RAIN):** Eliminates >99.98% of stateful SRAM Single-Event Upset (SEU) cross-section while preventing RowHammer (ARVALID=0) and PCIe Completion Timeouts via a 341-LUT per-ARID zero-completion responder.
- **Zero-LUT Model-Weight Anti-Exfiltration (thm27/thm29):** Hardware read-path wire-slicing clamps BF16/FP16 mantissas in 0 extra LUTs, erecting an 85× Le Cam minimax query-complexity wall against frontier model weight extraction.

## 4. TURNKEY COTS PROCUREMENT & RAPID CONTRACTING VEHICLES

| SKU / Track | Firm-Fixed-Price | Contracting Vehicle | Deliverables (Immediate Electronic / AWS F2 Delivery) |
|---|---|---|---|
| **SKU-1 (Eval Drop)** | **$9,850.00** | FAR 13.201 GPC Micro-Purchase | 45-Day 16nm xcvu47p Eval Drop, 30 Lean 4 Proofs, C11 BAR0 Driver, Python Digital Twin. |
| **SKU-2 (Full Pilot)** | **$24,950.00** | FAR 13.201 Lab GPC (≤$25k) | 90-Day Dual-AFI AWS F2 Share + 7nm ASAP7 Package + spectralc Compiler + k3s Operator. |
| **SKU-3 (Core License)** | **$95,000.00** | FAR Part 12 / 10 U.S.C. § 4022 OTA | Single-Program IEEE 1735 Encrypted EDIF Netlist, SVA VIP Suite, DO-254/RAIN Package. |

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